Transformer Meets GAN: Cloud-Free Multispectral Image Reconstruction via Multisensor Data Fusion in Satellite Images

Congyu Li, Xinxin Liu, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Cloud-free image reconstruction is of great significance for improving the quality of optical satellite images that are vulnerable to bad weather. When cloud cover makes it impossible to obtain information under the cloud, auxiliary data are indispensable to guide the reconstruction of the cloud-contaminated area. In addition, the areas that require continuous observation are mostly regions with complex features, which puts higher demands on the restoration of texture, color, and other details in data reconstruction. In this article, we propose a transformer-based generative adversarial network for cloud-free multispectral image reconstruction (TransGAN-CFR) via multisensor data fusion in satellite images. Synthetic aperture radar (SAR) images that are not affected by clouds are used as auxiliary data and paired with cloudy optical images into the generative adversarial network (GAN) generator. To take advantage of the deep-shallow features and global–local geographical proximity in remote sensing images, the proposed generator uses a hierarchical encoder–decoder structure, in which the transformer blocks adopt a nonoverlapping window multihead self-attention (WMSA) mechanism and a modified feedforward network (FFN) through depthwise convolutions and the gating mechanism. Besides, we introduce a triplet loss function specifically designed for cloud removal tasks to provide the generated cloud-less image with greater proximity to the ground truth. Compared with seven state-of-the-art deep-learning-based cloud removal models, our network can yield more natural cloud-free images with better visual performance and more accurate results in quantitative evaluation on the SEN12MS-CR dataset.**

***Index Terms*— Cloud removal, generative adversarial network (GAN), image fusion, optical image, synthetic aperture radar (SAR) image, transformer.**

I. Introduction

EARTH observation using remote sensing technology has attracted the widespread concern of researchers due to the launch and data release of increasingly more satellites. Remote sensing approaches have played a vital role in a variety of application scenarios, such as natural resource surveys, ecological environment protection, meteorological disaster prediction and emergency response, and national major project construction [1], [2], [3]. Among the massive remote sensing data, optical images have rich spectral information, strong interpretability, and good visual effect of ground objects, thus are crucial for researching and exploring the world. However, due to the passive imaging mode, optical images are prone to information degradation or missing when obscured by clouds, making it impossible to achieve comprehensive observation of ground objects [4], [5]. An investigation on the availability of Landsat enhanced thematic mapper plus (ETM+) satellite data [6] pointed out that nearly 35% of the Earth's surface is covered by clouds. Furthermore, another literature [7] counted the cloud mask of the moderate-resolution imaging spectroradiometer (MODIS) data and calculated that the cloud coverage on land is roughly 55% while the global cloud coverage is around 67%. The existence of large-scale and frequent cloud cover has a great adverse impact on the subsequent processing and application of optical satellite images, especially for scientific researches and project applications that rely on continuous observation of optical image time series [8].

Thick clouds often cause severe signal damage, so additional auxiliary data are essential to restore the missing information of underlying surface. Unlike optical remote sensing imaging, synthetic aperture radar (SAR) uses an active imaging approach to provide energy sources within the microwave range and receive the reflected signals of ground objects. It can achieve all-weather monitoring without being affected by atmospheric or weather conditions to ensure continuous observation of target areas. Due to easy access to SAR data, it is possible to solve the problem of removing thick clouds in optical images using SAR data as auxiliary information. Some optical image cloud removal methods combined with SAR have also been proposed in the past decades [9], [10], [11]. Nevertheless, due to the domain gap between SAR and optical image, the key issue is to effectively exploit the faculty in SAR data in cloud removal [12], [13]. Specifically, many existing cloud-free reconstruction methods only learn the pixel-level mapping relationship between optical and SAR images, neglecting the global–local information within the images. In addition, since

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the areas requiring continuous observation are mostly regions with complex surface features, such as vegetation or urban areas, it is also a challenge to generate cloud-free optical reconstruction results with more complete texture, realistic color, and detailed features in these areas. Finally, if the cloud removal task requires the reconstruction of all the bands in multispectral images, the computational cost of the model cannot be ignored.

To solve the above-mentioned challenges when removing clouds from multispectral remote sensing images, we propose a novel transformer-based generative adversarial network for cloud-free image reconstruction (TransGAN-CFR) by fusing SAR and optical images. The SAR images are used as auxiliary input to guide the global reconstruction because of its reliable ground object information. Inspired by the high modeling flexibility and strong generation ability of the generative adversarial network (GAN), we select GAN as the main structure of our framework, within which the generator and loss function are newly designed. To leverage the hierarchical encoder–decoder architecture's ability to integrate shallow and deep features, alongside the transformer network's process in preserving global and local features, we construct an encoder–decoder generative network with skip connections built on the transformer blocks. The transformer blocks incorporate a nonoverlapping window-based multihead self-attention (WMSA) mechanism and a feedforward network (FFN) in a double-flow gating structure with depthwise convolution to enhance feature extraction performance. Moreover, a specifically designed loss function is introduced to constrain the relationship among the generated image, real cloud-free image, and cloudy image. We compare the proposed TransGAN-CFR method with seven state-of-the-art deep learning based methods for cloud removal. The results show that the proposed method has remarkable improvement in visual and quantitative assessment. The ablation experiments demonstrate the effectiveness and necessity of special design for different modules.

Specifically, the contributions of this research can be summarized as follows.

1) We propose a novel SAR-assisted cloud-free image reconstruction model based on a GAN. The guidance of SAR images enables the reliable recovery of regions seriously affected by clouds and contributes to keeping rich texture details of the generated results. Unlike other cloud removal methods that only focus on the three visible bands of RGB, our algorithm considers each band of multispectral images and is devoted to restoring optical images with complete spectral information.
2) We design a hierarchical encoder–decoder generator network, which is composed of transformer blocks. The outer encoder–decoder network architecture combines shallow and deep features via downsampling, upsampling, and skip connections. Through nonoverlapping window multihead attention mechanism and FFN qualified with parallel double-branch gating and depthwise convolution structure in the transformer blocks, multiscale features and strengthened local context information can better learn the geographical proximity and reflect the global–local relationship in the reconstruction process at a small computing cost.
3) Considering the characteristics of image triplets in cloud-free image reconstruction tasks, we introduce triplet loss originally designed for cloud-free image reconstruction, to enhance the correlation between the generated image and the ground truth in aspects of texture, color, and other details during training. Triplet loss can allow the model to maintain more detailed features consistent with the real cloud-free image and confirm the reconstruction quality of the generated image, since its principle is to draw the generated image closer to the real cloud-free image but farther from the real cloudy image.

The remainder of this article is organized as follows. In Section II, we briefly overview the related work on cloud removal. Section III introduces the components of our proposed TransGAN-CFR method, including the overall framework, the generator architecture, the discriminator architecture, and the loss function. In Section IV, the experimental settings are provided, and the experimental results are shown and analyzed. The last section gives brief conclusions.

II. Related Work

To remove clouds from optical remote sensing images and restore the ground objects under the clouds, scholars have proposed various methods. These methods can be roughly divided into traditional methods and deep learning methods. According to the mechanism involved in data reconstruction, the traditional methods can be further divided into multispectral-based methods, multitemporal-based methods, and multisensor-based methods.

A. Traditional Methods

Multispectral-based methods are often applied to optical images that are only partially affected by clouds. In other words, the image still contains weak signals for recovering the affected surface information without using supplementary data [14]. Zhang et al. [15] developed a haze optimization transformation (HOT) method based on the analysis of visible band space for the detection and characterization of haze or thin cloud. Maalouf et al. [16] proposed an effective inpainting technique based on bandelet transform and multiscale geometric grouping to reconstruct cloud-contaminated images. Xu et al. [17] designed a thin cloud removal method using signal transmission principles and spectral mixture analysis based on the cloud distortion model. Concisely, the implementation of these methods usually depends on mathematical, physical, or geographical models. Due to the assumptions for determining the solution model, their application has some limitations, such as the inability to deal with large cloud-contaminated areas and sensitivity to shadows or bright objects.

Multitemporal-based methods restore cloud-contaminated images using cloud-free optical images obtained at other times as auxiliary information. In [18], a cloud-free image from another time was introduced as a guide to locate similar pixels based on the spatio-temporal Markov random fields

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(MRFs) global function. Cloud pixels were then replaced by similar pixels to restore missing information. Li et al. [19] replaced the cloud-covered area in the target image with the complementary region adjusted by the gradual radiometric measurement and further optimized the final recovery result via residual correction. These information cloning methods based on time correlation are prone to inaccurate restoration results when the land cover changes considerably in the reference temporal images. To improve the reliability of cloud removal, Wang et al. [20] provided an algorithm to recover underlying images from heavily cloud-contaminated images via temporally contiguous robust matrix completion. Xu et al. [21] introduced multitemporal dictionary learning (MDL) to cloud removal by combining the learned coefficients from the reference image and the dictionary from the target image. Ji et al. [22] presented a box-constrained model which separately represented the cloud components and the image components by a group sparsity function along the spectral dimension and the tensor rank based on singular value decomposition. Nevertheless, the model-driven multitemporal cloud removal methods, which depend on sparse representation, matrix completion, and other optimization-based techniques, often require complex parameter presetting and high time consumption.

As cloud-free data from the same platform or sensor at adjacent time are not always available, many researchers have focused on developing *multisensor-based methods* for cloud removal. In [23], the cloudy pixels were replaced by the cloud-free pixels in the same image whose location was speculated according to the nearest neighborhood relationship of the referenced SAR image. Li et al. [24] exploited the auxiliary images from two different sensors, the low-resolution optical image and the SAR image, to restore the cloud-covered regions in the high-resolution optical image, where dictionary group learning and nonlocal joint sparse coding were adopted to construct the relationship between images. In [25], inspired by the methods of spatial–temporal data fusion, the authors restored a cloudy Landsat-8 image from two low-spatial-resolution MODIS images and another Landsat-8 image using techniques of nonnegative matrix factorization and error correction. Generally speaking, ancillary images are easier to obtain from different sensors, and their acquisition time can be closer to the study time. However, due to the different imaging mechanisms of imaging sensors, the domain gap between multisensor data is an inevitable problem that harms the relevant feature extraction.

B. Deep Learning Methods

In recent years, deep learning has made significant progress in computer vision and has many successful applications in the field of image restoration [26], [27], [28], [29]. The data-driven deep learning algorithms specialize in extracting more comprehensive and accurate features and in mapping of complex relationships. Up to now, many outstanding deep learning models have been migrated and used successfully in remote sensing image processing tasks to break through the limitations of the traditional methods. For example, Wu et al. [30] embedded tiny UNet networks into the overall UNet backbone network for small target detection in infrared images, which effectively enhances the distinguishability of features. Hong et al. [31] introduced graph convolutional networks into hyperspectral image classification to strengthen the model's ability by capturing medium- to long-term spatial relationships. Yang et al. [32] detected changes in irregularly collected images using an unsupervised time–distance-guided convolutional recurrent network. Therefore, deep learning methods can better mine the in-depth features of multispectral, multitemporal, and multisensor data, so as to achieve superior effects in cloud removal [33]. Zhang et al. [34] proposed a deep learning framework based on spatial–temporal patch group, which used a global–local loss function for weighted aggregation and progressive iteration to reconstruct the overall cloud removal results. In [35], the authors provided a dual full convolution network (FCN) embedded with multiscale aggregation and channel attention mechanism, which realized cloud detection and cloud removal simultaneously through bitemporal images. A deep residual neural network fusing Sentinel-1 and Sentinel-2 satellite images to recover the cloud-contaminated images was presented in [36].

Recently, GANs [37] have become increasingly popular and shown effective performance in many fields due to the balance strategy between the generator and the discriminator. In the relevant research of cloud-free image reconstruction, the methods based on GAN also account for a considerable part. For example, Enomoto et al. [38] exploited conditional GAN (cGAN) to remove cloud from multispectral images. A Cloud-Gan [39] was proposed based on cyclic consistent GANs to learn the relationship between cloudy image and clear image. Another deep generation network was proposed in [40] considering the spatial attention mechanism and the relevant loss. In addition, on the basis of GAN framework, Xu et al. [41] presented the AMGAN method with an optimized generator, which is composed of an attentive recurrent network, an attentive residual network, and a reconstruction network. The attentive recurrent network can precisely generate the attention map of the cloud through residual blocks and long short-term memory (LSTM) mechanism. Among the cloud removal method based on GAN, there are also many methods requiring auxiliary data like SAR images. Darbaghshahi et al. [11] designed two GANs to remove clouds, in which the former translated SAR images into primary cloudless optical images and the latter removed clouds using the translated images from previous GAN and the real cloud-contaminated images. However, due to the domain gap between SAR images and optical images, it is a challenge to ensure the reliability and quality of the generated optical image [12]. In this case, SAR–optical fusion is introduced to guide the reconstruction. Grohnfeldt et al. [42] first designed the cGAN network with the integration of SAR–optical fusion module to generate cloud-less optical images with the assistance of SAR data. Xiao and Lu [43] used the SAR image to inpaint the edge map of the cloud-contaminated image, and then used both the images to facilitate the reconstruction of the optical image.

Inspired by the performance of the GAN architecture, we propose the TransGAN-CFR framework described in

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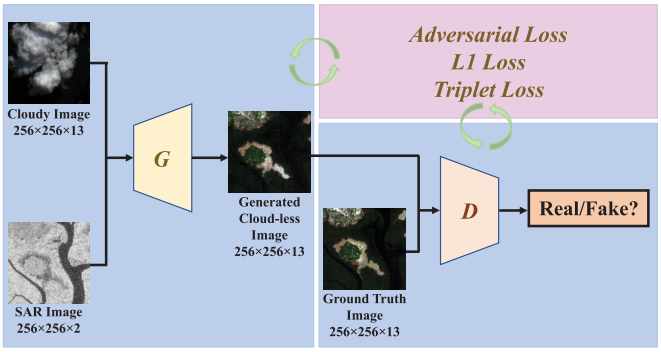


Fig. 1. Overview of the proposed TransGAN-CFR.

Section III, which is able to restore the cloud-covered optical images with complete spectral information and texture details with the assistance of SAR data.

III. Methods

A. Overview of the Proposed Framework

The proposed TransGAN-CFR is developed based on the GAN, which has three important components: generator, discriminator, and loss function. The main purpose of the GAN can be summarized as two balanced goals: to minimize the objective function of the generator and maximize that of the discriminator. Specifically, the purpose of the generator is to synthesize fake images that are similar to the ground truth to treat the discriminator, while the goal of the discriminator is to distinguish fake generated images from real images for revising the training direction of the generator. In this way, the discriminator and generator get a tradeoff to reach the Nash equilibrium, so that a high-quality generative network can be trained.

As presented in Fig. 1, the proposed method first uses cloudy optical and SAR image pairs as the input of the generator to generate cloud-free synthetic optical images and then feeds the generated and real cloud-free optical images (i.e., ground truth) into the discriminator for judgment. After reasonable and complete training, we can predict the cloud-contaminated regions in optical images through the convergent generative network. The generator is a hierarchical encoder–decoder network, in which the transformer blocks are used to capture self-attention information and spatial local context information from multiscale features with low computational cost. The discriminator is a basic convolution neural network (CNN) for binary classification. The total loss function is the weighted sum of adversarial loss function $\mathcal{L}_{\text{cGAN}}$ in the conditional GAN, L1 loss function $\mathcal{L}_{L1}$, and the triplet loss function $\mathcal{L}_{\text{triplet}}$. Specific designs of our network architecture are described in the rest of this section.

B. Generative Network

The generator learns the features of the input data through an elaborately designed depth architecture to generate fake cloud-less optical images. UNet, originally designed for medical image segmentation, provides a powerful U-shaped encoder–decoder structure [44], which includes the feature extraction parts on the left (i.e., encoder stages) and the feature fusion parts on the right (i.e., decoder stages). Owing to the U-shaped structure with skip connections, UNet can retrieve shallow features through feature concatenation with deep features, which is in line with the principle of combining global and local information in image restoration tasks. Since the UNet structure can make the generated results more comprehensive and complete, it is then modified for training and generating images in GAN.

Note that the UNet architecture uses a CNN which has a restricted receptive field. It is necessary to stack the CNN at multiple stages to acquire global information. Unfortunately, as the network goes deeper, the amount of valid information will be exhausted. Namely, global information from deeper CNN layers is still limited. Compared with the CNN block, transformer is more flexible and effective to capture global dependencies through self-attention mechanism. Moreover, multihead self-attention (MSA) can project the attention map into different dimensions and obtain more feature expression in various subspaces. We intend to train an excellent cloud-free image generation network by benefiting from the shallow–deep feature captured from the encoder–decoder structure and the global–local awareness of the transformer.

As shown in Fig. 2, we adopt the transformer blocks as the basic blocks and use skip connections between the encoder and decoder stages to form the overall U-shaped architecture of the generator. Each encoder/decoder stage consists of two contiguous transformer blocks and a sampling block. Given that the input feature map in a transformer block is $\mathcal{F}_{\text{in}}$ and the output feature map is $\mathcal{F}_{\text{out}}$, the basic architecture of the transformer block in Fig. 2(a) can be mathematically expressed as follows:

$$\begin{aligned}\mathcal{F}_{\text{mid}} &= \text{WMSA}(\text{LN}(\mathcal{F}_{\text{in}})) + \mathcal{F}_{\text{in}}\\ \mathcal{F}_{\text{out}} &= \text{FFN}(\text{LN}(\mathcal{F}_{\text{mid}})) + \mathcal{F}_{\text{mid}}.\end{aligned} \tag{1}$$

Here, WMSA and FFN are the core modules in the transformer architecture. LN refers to the operation of layer normalization [45]. $\mathcal{F}_{\text{mid}}$ represents the output of the WMSA module. Next, the WMSA and FFN modules of our transformer blocks are introduced in detail, respectively.

1) WMSA Module: Considering geographical proximity, multiscale features, and high complexity of remote sensing images, we perform nonoverlapping window-based MSA instead of global self-attention of the standard vision-transformer [27], [46] in the proposed transformer block.

Let $\mathcal{F} \in \mathbb{R}^{W\times H\times C}$ be the feature map with a spatial size of $W \times H$. We split the map $\mathcal{F}$ into nonoverlapping patches through an $M \times M$ window. Then the MSA operation is performed on each flattened and transposed split feature map patch $\mathcal{F}_i \in \mathbb{R}^{M\times M\times C}$ from the ith window $\mathbf{W}_i$. Suppose k is the head number and $\{d_k = C/k, k \in K\}$ is the head dimension of the MSA operation. The WMSA process which is shown in Fig. 2(b) can be formulated as follows:

$$\begin{aligned}\mathcal{F} &= \{\mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_N\}, \quad N = \frac{WH}{M^2}\\ \hat{\mathcal{F}}^k &= \left\{\hat{\mathcal{F}}^k{}_1, \hat{\mathcal{F}}^k{}_2, \ldots, \hat{\mathcal{F}}^k{}_N\right\}, \quad N = \frac{WH}{M^2}\end{aligned} \tag{2}$$

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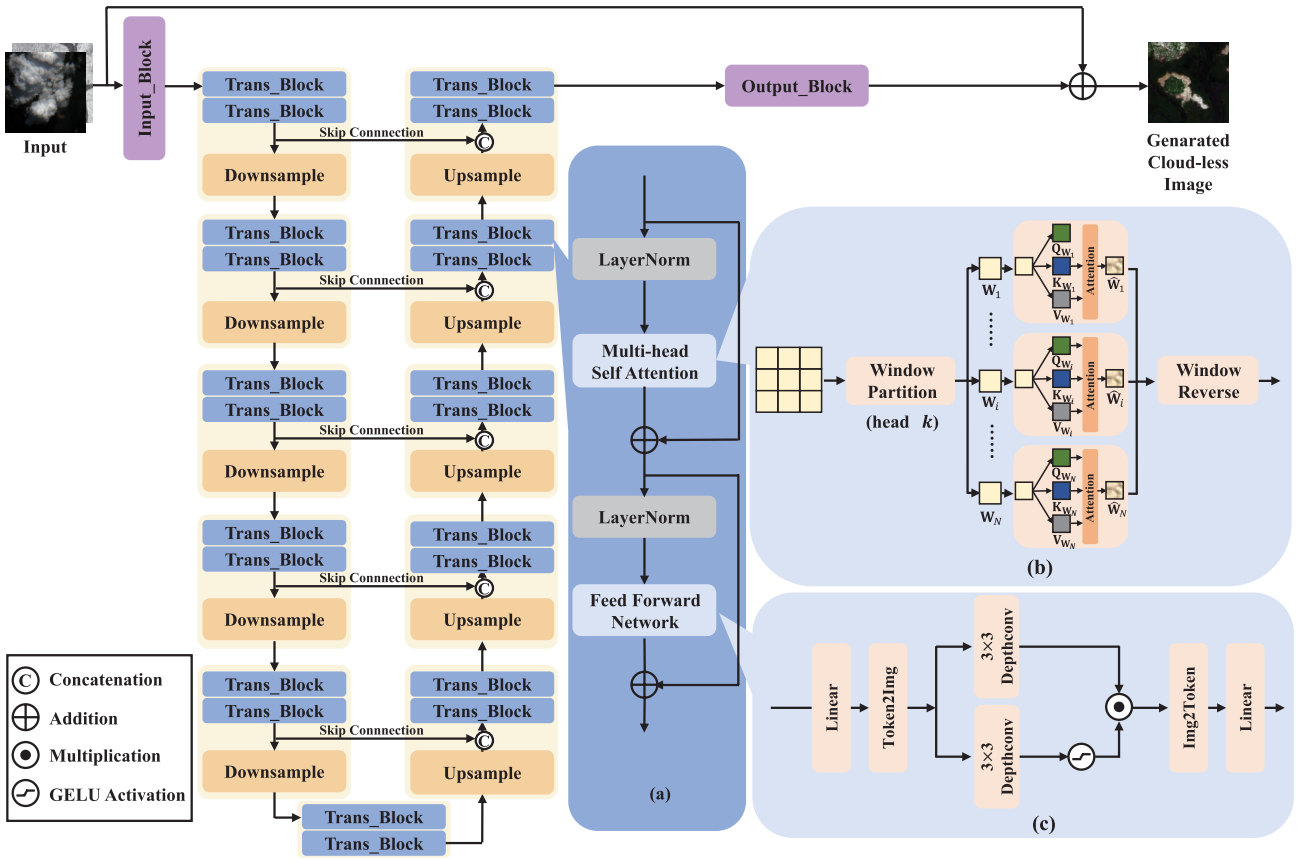


Fig. 2. Architecture of the generative network in TransGAN-CFR. The generator's overall pipeline is a symmetric U-shaped encoder–decoder structure with skip connections between the encoding layer and the decoding layer based on transformer blocks. (a) Transformer block. (b) WMSA module. (c) FFN module.

where $\hat{\mathcal{F}}^k$ refers to the attention result of the kth head. The attention of $\hat{\mathcal{F}}_i^k$ can be calculated based on the relative position embedding strategy [47] as follows:

$$\begin{aligned}\left(\mathbf{Q}_{\mathbf{W}_i}^k, \mathbf{K}_{\mathbf{W}_i}^k, \mathbf{V}_{\mathbf{W}_i}^k\right) &= \text{PositionEmbedding}(\mathbf{W}_i)_k \\ \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{SoftMax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{B}\right)\mathbf{V} \\ \hat{\mathcal{F}}_i^k &= \text{Attention}\left(\mathbf{Q}_{\mathbf{W}_i}^k, \mathbf{K}_{\mathbf{W}_i}^k, \mathbf{V}_{\mathbf{W}_i}^k\right), \\ & \quad i = 1, \ldots, N \end{aligned} \tag{3}$$

where $\mathbf{Q}_{\mathbf{W}_i}^k$, $\mathbf{K}_{\mathbf{W}_i}^k$, and $\mathbf{V}_{\mathbf{W}_i}^k$ represent the *query*, *key*, and *value* tokens in the ith window when the head number is k, respectively. $\mathbf{B}$ represents the relative position bias. Thus, the final WMSA result is the concatenated and projected linear attention results of all the heads. The WMSA strategy transforms input feature maps into multiple subspaces through multiple heads, which allows the model to focus on the information at different angles.

Compared with global self-attention, WMSA can significantly reduce the calculation cost. With regard to feature maps represented as $\mathcal{F} \in \mathbb{R}^{W\times H\times C}$, the computational complexity of global self-attention is $O\left(H^2W^2C\right)$, while that of nonoverlapping WMSA is $O\left((HW/M^2)M^4C\right)$, i.e., $O\left(HWM^2C\right)$. Generally speaking, this strategy greatly reduces the computational cost. Since we use multichannel input data and design the main body of the generator into a deep encoder and decoder structure, this WMSA module can greatly improve the efficiency of training and convergence.

2) Feedforward Network: As an essential structure of the transformer, the FFN can extract and learn more abstract features to promote the expression ability of the model by mapping data to high-dimensional space and then to low-dimensional space. Noting the contribution of spatial information, i.e., geographical neighborhood relationships, for image restoration [48], the FFN for our task ought to focus on the complementary local spatial information and reduce the coupling between feature channels. Inspired by [49], we adopt the depthwise convolution and gating mechanism to build the FFN module, which exactly meets our needs and expectations.

As displayed in Fig. 2(c), the FFN module first doubles the dimension of the input tokens by a linear projection layer and reshapes the tokens into feature maps. Then, the feature maps are divided into two parallel branches to capture local information via depthwise convolution (DepthConv) blocks, where one branch is elementwise multiplied by the other branch activated by the nonlinear GELU. Finally, the feature maps are flattened back to tokens and the dimensions are rearranged to be consistent with the input.

The mode of DepthConv performed on each separate channel can decouple the correlation between convolutional layer channels to ensure that the output result is less disturbed by the correlation between feature channels [50]. The low-coupling feature maps obtained by different convolution paths from

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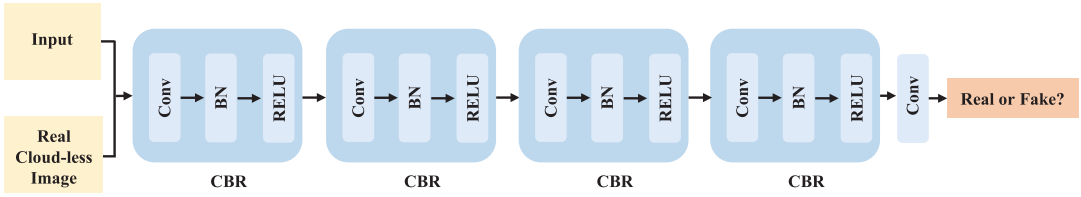


Fig. 3. Architecture of the discriminative network in TransGAN-CFR. The discriminator consists of four **CBR**s and one convolutional layer, where **C** represents convolutions, **B** refers to the batch normalization, and **R** means the activation function ReLU.

different feature subspaces can obtain more complete, comprehensive, and complementary information to represent the image. Moreover, the dual-flow gating mechanism enhances the spatial information in each convolution channel by filtering the information flowing through the network. This leads to a complementation of spatial information between channels, resulting in increased accuracy and efficiency. Thus, with the support of DepthConv operation and gating mechanism, the FFN can promote the extraction of enhanced local context in spatial information in a lightweight way with less computational parameters and cost.

C. Discriminative Network

The discriminator is used to judge whether the input image is real or fake by the output discrimination possibility. Inspired by the work in [51], we build the discriminator as shown in Fig. 3. The discriminator structure includes four **CBR**s and one convolutional layer, where **C** represents convolutions, **B** refers to the batch normalization, and **R** means the Leaky ReLU activation function. In detail, the convolution kernel of each **CBR** is 4×4 and the stride is 2, while the kernel of the last convolutional layer is 3×3 and the stride is 1. Four CBRs expand the input data into 32-D, 64-D, 128-D, and 256-D features, respectively. The last convolutional layer transforms the 256-D feature into a 1-D flag that can be used to indicate whether the input image is a real cloud-free image or a generated fake image.

D. Loss Function

The loss function $\mathcal{L}_{\text{TransGAN}-\text{CFR}}$ of our proposed method is composed of adversarial loss function $\mathcal{L}_{\text{cGAN}}$ in the conditional GAN, L1 loss function $\mathcal{L}_{L1}$, and the triplet loss function $\mathcal{L}_{\text{triplet}}$, which can be denoted as

$$\begin{aligned}\mathcal{L}_{\text{TransGAN-CFR}} = &\arg\min_{G}\max_{D}\mathcal{L}_{\text{cGAN}}(G,D)\\ &+\lambda_{L1}\mathcal{L}_{L1}(G)+\lambda_{\text{triplet}}\mathcal{L}_{\text{triplet}}\end{aligned} \tag{4}$$

where λ_{L1} and λ_{triplet} are the weight coefficients of $\mathcal{L}_{L1}(G)$ and $\mathcal{L}_{\text{triplet}}$ in the loss function, respectively.

According to the goal of GAN, the generator G is committed to finding a function to produce the generated image that can confuse the discriminator D. Namely, the generator G aims at minimizing the difference between the generated image and ground truth, while the discriminator D aims at maximizing that difference. The loss of conditional GAN in our model is formulated as

$$\begin{aligned}\mathcal{L}_{\text{cGAN}}(G,D) = &E_{x,y\sim p_{\text{data}}(x,y)}\left[\log D(x,y)\right]\\ &+E_{x\sim p_{\text{data}}(x)}\left[\log(1-D(x,G(x)))\right]\end{aligned} \tag{5}$$

where x refers to the input data including the cloudy image and SAR image. y refers to the real cloud-free optical image. $p_{\text{data}}(x)$ represents the distribution of x. $G(x)$ is the generated cloud-free image from the generator. $D(\cdot,\cdot)$ is the proximity probability of two images calculated by the discriminator.

To ensure the quality of the generated image with less blurring, the L1 loss function $\mathcal{L}_{L1}(G)$ is added as follows:

$$\mathcal{L}_{L1}(G) = \frac{1}{CHW}\sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W}\lambda_c\left|y^{(w,h,c)}-G(x)^{(w,h,c)}\right|_1 \tag{6}$$

where C, H, and W refer to the size of the optical image in 3-D. (w,h,c) represents the location of a pixel. λ_c represents the balance factor of each band in the multispectral image for the loss calculation. $\mathcal{L}_{L1}(G)$ can determine the difference between each reconstructed pixel $G(x)^{(w,h,c)}$ and its corresponding real cloud-less pixel $y^{(w,h,c)}$.

Triplet loss [52] is calculated based on triples of anchor sample $\mathbf{x}_i^a$, positive sample $\mathbf{x}_i^p$, and negative sample $\mathbf{x}_i^n$. The goal is to ensure the anchor $\mathbf{x}_i^a$ is closer to positive sample $\mathbf{x}_i^p$ but farther away from the negative sample $\mathbf{x}_i^n$, which satisfies the inequality

$$\left\|\mathbf{x}_i^a-\mathbf{x}_i^p\right\|_2^2+\alpha<\left\|\mathbf{x}_i^a-\mathbf{x}_i^n\right\|_2^2 \quad \forall\left(\mathbf{x}_i^a,\mathbf{x}_i^p,\mathbf{x}_i^n\right)\in\mathbb{T} \tag{7}$$

where $\mathbb{T}$ refers to the set of all possible triplets and i is the order number of the calculated triplets. α is an enforced constant. For convenience of model constraint, the inequality can then be rewritten as triplet loss $\mathcal{L}_{\text{triplet}}$ as follows:

$$\mathcal{L}_{\text{triplet}}=\sum_{i}^{N}\left[\left\|f\left(\mathbf{x}_i^a\right)-f\left(\mathbf{x}_i^p\right)\right\|_2^2-\left\|f\left(\mathbf{x}_i^a\right)-f\left(\mathbf{x}_i^n\right)\right\|_2^2+\alpha\right]_+ \tag{8}$$

where $f(\cdot)$ represents the embedding method for the required feature, and N is the total number of triplets. $[Z]_+$ represents the operation to take the higher value between 0 and Z. Therefore, the output of triplet loss aims to penalize the case of $\|f(\mathbf{x}_i^a)-f(\mathbf{x}_i^p)\|_2^2>\|f(\mathbf{x}_i^a)-f(\mathbf{x}_i^n)\|_2^2$.

For the task of cloud-free image reconstruction, the anchor samples $\mathbf{x}^a$ are the generated cloud-less optical images, the positive sample $\mathbf{x}^p$ are the real cloud-less optical images, and the negative samples $\mathbf{x}^n$ are the cloudy optical images. Thus, the triplet loss $\mathcal{L}_{\text{triplet}}$ in the proposed model aims to

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minimize the difference between the generated image and the real cloud-free image and maximize the difference between the generated image and the real cloudy image, whose intention is to guide the training direction of GAN to avoid generating poor-quality fake images for the sake of the reconstruction reliability. In addition, the use of triplet loss can also help the model converge more efficiently.

IV. Experiments and Results

A. Experimental Settings

1) Dataset Preparation: We used SEN12MS-CR [8] dataset to train and test all the methods, which was built from downloadable Sentinel satellite data provided by the Copernicus program. The dataset including triplets of Sentinel-2 optical images, Sentinel-1 SAR images, and land cover maps from MODIS is specially created for the development of deep learning cloud removal models. Samples in each triplet are orthorectified, geo-referenced, and close to each other in acquisition time. In this article, we adopted the grouped cloudy Sentinel-2 images, cloud-less Sentinel-2 images, and Sentinel-1 SAR images for experiments.

The size of each image patch is 256×256. Multispectral images of Sentinel-2 have 13 bands. The Sentinel-1 SAR image contains two polarization channels obtained in the IW mode (VV and VH). Before model training, we preprocessed the data in the same way as [12]. The pixel values of optical images are within the range of $[0, 10\,000]$, while the data values of VV and VH channels in SAR data are clipped to the range of $[-25, 0]$ and $[-32.5, 0]$, respectively. All the data were then normalized between 0 and 1 to facilitate the accuracy and speed of network training. For the reliability of the experiment, we selected two regions of interest (ROIs) with different geographical locations and seasons from SEN12MS-CR. ROI 1 mainly contains samples of various vegetation and wetlands in Northern Red Sea Karora, Eritrea, captured in spring. ROI 2 covers the town scenarios in Flemish Region East Flanders, Belgium, obtained in summer. Each ROI consists of 784 samples, 64% of which were used for training, 16% for validation, and the remaining 20% for testing. There are far more thick cloud covered patches than thin cloud or cloudless ones in these two ROIs. For better quantitative evaluation, the cloud coverage condition of these two ROIs is recorded by MSE, SAM, PSNR, and SSIM measures between the real cloudy and the cloud-free images (denoted by raw in Table I).

2) Evaluation Metrics: Four full-reference indices including spectral angle mapper (SAM), mean-squared error (MSE), peak signal-to-noise ratio (PSNR), and structural similarity index measure (SSIM) were used to quantitatively compare the cloud removal performance of different methods. For ease of illustration, the cloud-less image (i.e., ground truth) is denoted by $\hat{\mathcal{X}} \in \mathbb{R}^{W\times H\times C}$, while the reconstructed image is denoted by $\mathcal{X} \in \mathbb{R}^{W\times H\times C}$.

SAM [53] measures the quality of reconstruction by calculating the arccosine value between the spectral vector $\hat{x}_i$ and $\mathbf{x}_i$ at each pixel i in the reference image and the generated image as follows:

$$\mathrm{SAM}\left(\hat{x}_i, \mathbf{x}_i\right) = \arccos\left(\frac{\hat{x}_i{}^{\top}\mathbf{x}_i}{\|\hat{x}_i\|_2\|\mathbf{x}_i\|_2}\right) \tag{9}$$

where $\|\cdot\|_2$ represents the Euclidean norm. SAM is measured by degree, and its value can reflect the spectral distortion of the reconstructed image. The smaller the absolute SAM value, the higher the spectral quality of the reconstructed image. The overall SAM value of an image is calculated from the average SAM value of all the pixels.

MSE computes the error between the target image $\mathcal{X}$ and the ground truth $\hat{\mathcal{X}}$. A small MSE value indicates that the difference between two comparative images is small. The MSE of the kth band in an image is defined as

$$\mathrm{MSE}\left(\hat{X}_k, X_k\right) = \frac{\left(\hat{X}_k - X_k\right)^2}{W \cdot H}. \tag{10}$$

PSNR is calculated via the ratio of the maximum signal value in the reference image $\hat{\mathcal{X}}$ to the residual error between the generated image $\mathcal{X}$ and the reference image. Higher PSNR value indicates better spatial recovery quality. Its calculation formula is given as follows:

$$\mathrm{PSNR}\left(\hat{X}_k, X_k\right) = 10 \cdot \log_{10}\frac{\mathrm{MAX}\left(\hat{X}_k\right)^2}{\mathrm{MSE}\left(\hat{X}_k, X_k\right)}. \tag{11}$$

The SSIM can reflect the structural similarity of the generated image $\mathcal{X}$ and the reference image $\hat{\mathcal{X}}$, which can be computed via

$$\mathrm{SSIM}\left(\hat{X}_k, X_k\right) = \frac{\left(2\mu_{\hat{X}_k}\mu_{X_k} + C_1\right)\left(2\sigma_{\hat{X}_k X_k} + C_2\right)}{\left(\mu_{\hat{X}_k}^2 + \mu_{X_k}^2 + C_1\right)\left(\sigma_{\hat{X}_k}^2 + \sigma_{X_k}^2 + C_2\right)} \tag{12}$$

where $\mu_{\hat{X}_k}$ and μ_{X_k} represent the mean value of the kth band in $\hat{\mathcal{X}}_k$ and $\mathcal{X}_k$, respectively. $\sigma_{\hat{X}_k}$ and σ_{X_k} represent the standard deviation of the kth band in $\hat{\mathcal{X}}_k$ and $\mathcal{X}_k$, respectively. $\sigma_{\hat{X}_k X_k}$ represents the covariance of the kth band in image $\hat{\mathcal{X}}$ and $\mathcal{X}$. The value range of SSIM index is between 0 and 1. The closer the SSIM value is to 1, the more similar the two images are, and the better the structure recovery will be. In addition, the final MSE, PSNR, and SSIM value of multichannel images in this article were calculated by averaging the values taken at each band.

3) Implementation Details: Our training and testing experiments were executed on a Windows 11 operating system with a 12th Gen Intel[1] Core(TM) i9-12900K CPU at 3.20 GHz and an NVIDIA GeForce RTX 3090 GPU. The deep learning algorithm TransGAN-CFR was implemented by Python (version: 3.7) language with the PyTorch framework (version: 1.11.1). During training, according to the previous work and experience, the batch size was set to 1, the training epoch was set to 200, and the learning rate was set to 7×10^{-5}. The ADAM optimizer was adopted with an exponential moving average decay of 0.999.

To balance the accuracy and efficiency of model training, the encoder/decoder stages' number K_S of the symmetric

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TABLE I
QUANTITATIVE EVALUATION OF ALL THE TEST METHODS (THE BEST RESULT IS SHOWN IN BOLD)

Method	Input Data		ROI 1				ROI 2				FLOPs	Params
	Optical	SAR	MSE↓	SAM↓	PSNR↑	SSIM↑	MSE↓	SAM↓	PSNR↑	SSIM↑		
raw			0.046904	14.7504	21.6264	0.2571	0.067741	15.0413	16.7512	0.1771		
SpAGAN [40]	✓		0.002282	8.3135	33.2373	0.6479	0.001859	7.6755	33.4067	0.6716	17.55 GMac	225.06 k
AMGAN [41]	✓		0.001813	6.6801	35.0433	0.7043	0.001368	6.4371	34.6711	0.7224	50.05 GMac	249.79 k
Pix2pix [51]	✓		0.001724	7.5033	33.4226	0.6766	0.000926	5.5793	35.0505	0.8215	61.0 GMac	11.45 M
McGAN [38]	✓		0.005826	5.5112	36.7594	0.8324	0.000507	4.1765	38.5232	0.9006	72.27 GMac	54.43 M
SAR-Opt-cGAN [42]	✓	✓	0.000479	4.2102	39.5045	0.8927	0.000242	2.7839	39.1252	0.9283	91.68 GMac	218.03 M
Dsen2cr [36]	✓	✓	0.000245	3.1877	40.5030	0.9325	0.000216	2.7056	40.6891	0.9516	1241.48 GMac	18.94 M
Glf-CR [12]	✓	✓	0.000159	2.6646	42.9111	0.9556	0.000155	2.3221	42.3054	0.9650	244.63 GMac	14.83 M
TransGAN-CFR	✓	✓	**0.000149**	**2.2053**	**44.3781**	**0.9633**	**0.000139**	**2.0053**	**44.2738**	**0.9709**	20.99 GMac	74.43 M

hierarchical network in the generator was set to 5. The first four encoder stages were composed of two transformer blocks and a downsampling block that doubled the feature dimension, while the sampling block in the last encoder stages kept the feature dimension unchanged. At the bottom of the U-shaped structure, there were two connected transformer blocks to transmit the encoded features to the decoding stages. Specifically, for each stage's transformer block, the window size was set to 8 and the attention head number was set to [1, 2, 4, 8, 16, 32, 32, 16, 8, 4, 2], respectively. The number of CBRs in the discriminator was set to 4. As for the total loss function, λ_{L1} was set to 100, λ_{triplet} was set to 1, and λ_c was set to 1. To simplify parameters, α in λ_{triplet} was set to 0. When conducting comparative experiments, each comparison method adopted the same open code and parameter settings as the published paper. All the methods were trained for 200 epochs. To avoid unfair comparison caused by input data, the same training, validation, and testing samples were used.

B. Experimental Results

We compared our proposed method with seven state-of-the-art deep learning methods for cloud removal, including methods using a single cloudy optical image and methods with SAR data assistance. The former category includes SpAGAN [40], AMGAN [41], Pix2pix [51], and McGAN [38], while the latter category includes SAR-Opt-cGAN [42], DSen2-CR [36], and Glf-CR [12]. SpAGAN [40] focuses on the identification and reconstruction of cloud-contaminated regions in RGB images using local-to-global spatial attention. It is developed based on the GAN framework whose generator is a spatial attention network (SPANet) [54] with four spatial attentive blocks. AMGAN [41] is a cloud removal method proposed for Landsat-7 images, and its overall architecture is also attention-based GAN. Pix2pix is an image-to-image translation method based on conditional adversarial networks, which is able to learn the mapping from the input image to the output image under the constraint of the loss function. This work was originally designed for tasks including synthesizing, restoring, and colorizing natural photographs. In experiments, we extended the input of the Pix2pix method with multispectral data for comparison. McGAN [38] is a cGAN using UNet (with eight encoder–decoder stages) as the generator. It exploits the near-infrared bands as prior knowledge to remove clouds in four-band multispectral images. In this case, only optical multispectral images were used as the input for SpAGAN, AMGAN, Pix2pix, and McGAN. As for another group, SAR-Opt-cGAN [42] uses a cGAN architecture similar to the Pix2pix method with a bottleneck layer of 1024 features. In contrast, DSen2-CR [36] and Glf-CR [12] do not use the GAN structure. Specific difference between DSen2-CR and Glf-CR is that the former adopts a deep residual neural network developed from [55] and [56], while the latter build an SAR-guided two-stream network composed of global context interaction blocks and local feature compensation blocks. Since SAR-Opt-cGAN, DSen2-CR, and Glf-CR are cloud removal networks based on SAR and optical images, their input conditions are the same as the proposed method.

The quantitative evaluation results of the proposed TransGAN-CFR method and compared methods are presented in Table I. The quantitative results in the first row, denoted as *raw*, calculates a measure between the real cloudy image and the real cloud-free image, which can be used as the baseline for quantitative comparison of each method. In addition, the computational complexity of methods is also presented. Due to different frameworks used in comparative methods, from a fairness perspective, the FLOPs and learnable parameters from the generator module of each method are used for comparison. The number of channels for input data should follow the original method described by the author, as shown in Table I. More specifically, the numbers of input data channel are 13 for methods that only use optical images and 15 for methods that simultaneously use optical and SAR images. As shown in Table I, the proposed TransGAN-CFR outperforms the compared methods significantly in four quantitative evaluation indices at a relatively reasonable computational cost. The considerable improvement of indices indicates the validity of our model.

To demonstrate the effectiveness of the proposed method, the visual reconstruction results of all the methods of five scenes with different cloud coverage are provided in Fig. 4. Obviously, SpAGAN can restore certain ground objects in low cloud coverage images through a spatial attention architecture that involves a two-round, four-directional recurrent neural networks. However, the detail after reconstruction is insufficient to identify specific objects. AMGAN is able to generate accurate heat maps of clouds through the attention mechanism, but the reconstruction is not ideal because of undesirable artifacts. What is worse, SpAGAN and AMGAN are almost invalid when thick clouds completely cover the entire image, and the results only recover the basic hue and outline information of the ground objects under the cloud. Pix2pix is slightly better

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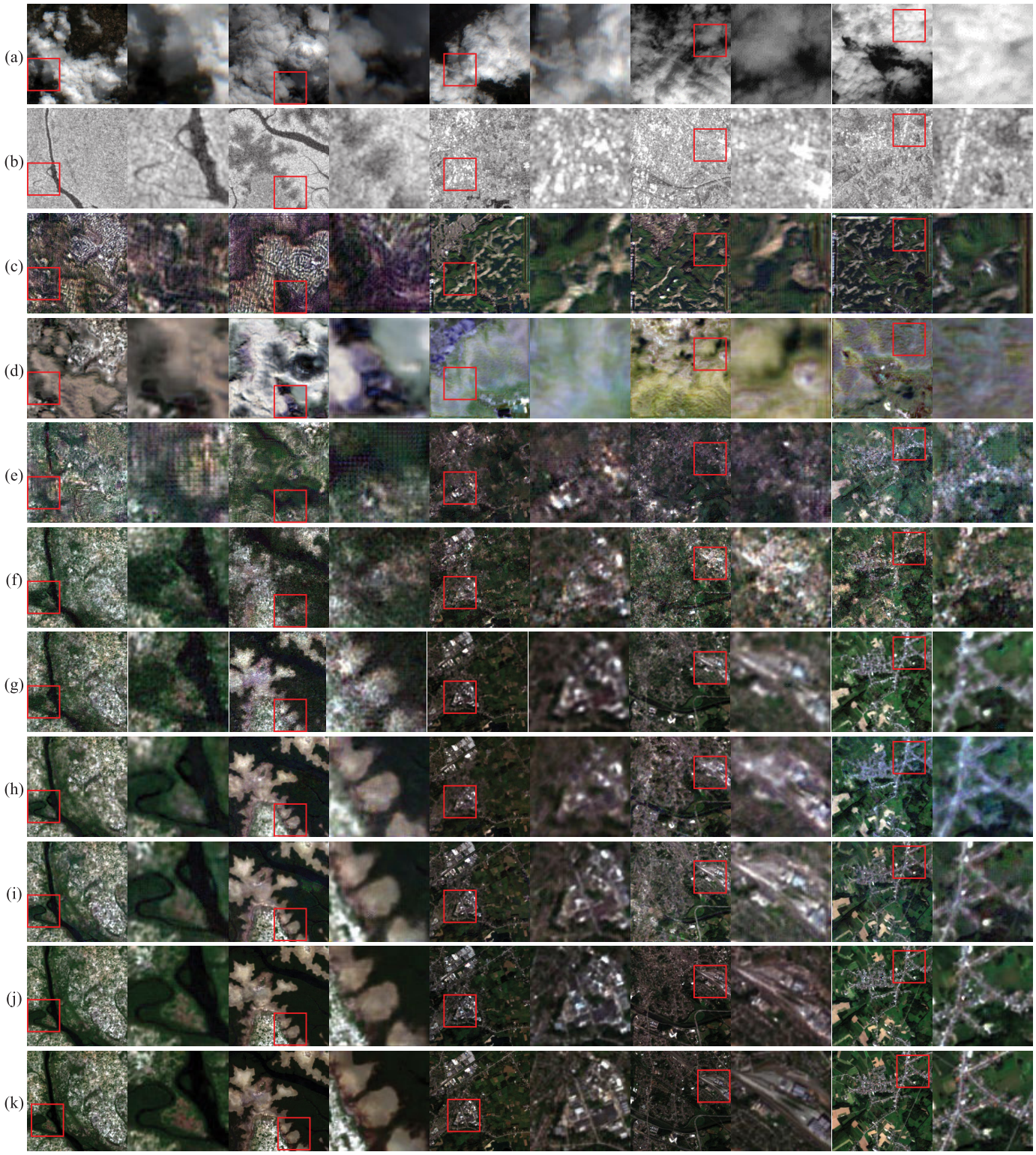


Fig. 4. Visual performance of all the test methods. The first two samples come from ROI 1, and the last three samples come from ROI 2. Even columns give the close-ups of the red boxes in odd columns. (a) Cloudy images (shown in RGB bands). (b) SAR images (shown in the VH channel). (c) SpAGAN [40]. (d) AMGAN [41]. (e) Pix2pix [51]. (f) McGAN [38]. (g) SAR-Opt-cGAN [42]. (h) DSen2-CR [36]. (i) Glf-CR [12]. (j) TransGAN-CFR. (k) Real cloud-free images.

than the first two methods in recovering ground objects under thick clouds, but its reconstruction results exhibit obvious noise and "block effects." To some extent, McGAN can restore more complete spectral and structural information of ground objects in thick cloud areas compared with the aforementioned methods. However, the severe noise and obvious color distortion in the results cannot be ignored. SAR-Opt-cGAN uses the similar generator as Pix2pix and adopts SAR images as

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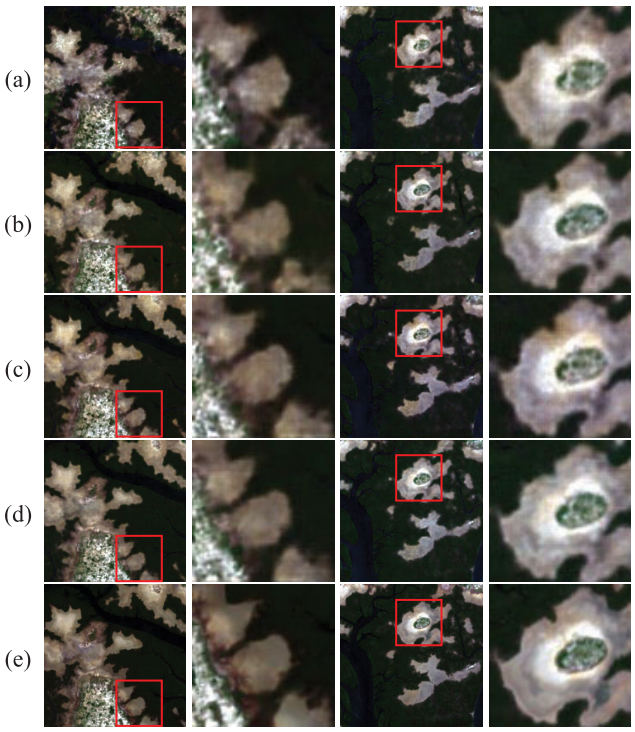


Fig. 5. Visual display of the effectiveness of SAR image and triplet loss in ablation study using two scenes as examples. Even columns give the close-ups of the red boxes in odd columns (a) w/o SAR nor triplet loss, (b) w/o SAR, (c) w/o triplet loss, (d) TransGAN-CFR, and (e) real cloud-free images.

additional input, so its results seem better when the cloud coverage is high. DSen2-CR can restore cloudy images but its results have obvious color distortion and oversmoothing problems, which leads to poor restoration performance for fine texture details. Glf-CR can reconstruct the image with better color and texture details by integrating SAR features with optical features, making it the most effective method among all the comparative methods.

For the proposed TransGAN-CFR method, it achieves the highest quantitative results and the best visual performance in tasks with different cloud coverage ratios in different regions (as shown in Table I and Fig. 4) because of the introduction of complementary information from SAR data and the utilization of hierarchical encoder–decoder structure combining transformer blocks to achieve deep–shallow feature extraction and global–local context interaction. According to the quantitative results of these methods in Table I, the ranking of all the methods is not affected by the location and season of training data, which can verify the reliability and stability of the proposed method.

C. Ablation Experiments

In this section, we conducted ablation experiments to analyze the effectiveness and necessity of each part in model design and validate the network module selection of the proposed method. The quantitative analysis of ablation experiments is based on the dataset of ROI 1.

1) Effectiveness of SAR Image and Triplet Loss: To determine whether the involved SAR data and triplet loss improve the network performance, we conducted an ablation study on the effectiveness of these two components. Specifically, the network without input SAR nor triplet loss is recorded as *w/o SAR nor triplet loss*, the network without input SAR is recorded as *w/o SAR*, and the network without triplet loss is recorded as *w/o triplet loss*. The corresponding qualitative and quantitative results are shown in Fig. 5 and Table II. Specifically, the first and second rows in Fig. 5 show that without SAR input, the generated results tend to exhibit blurred or fragmented boundaries. However, when SAR is included, the network is able to reconstruct texture information, even in regions that are heavily obscured by thick clouds. This indicates that the complementary information contained in SAR data can truly contribute to cloud removal tasks in optical images, which provides momentous assistance to restore the thick cloud-covered regions. From the third row in Fig. 5, it can be inspected that the network with triplet loss tends to produce results with more similar spectral information as the ground-truth images. Namely, triplet loss is appropriate and beneficial for the cloud removal task. The MSE, SAM, PSNR, and SSIM measures in Table II also indicate that the addition of SAR and triplet loss has positive impacts on network performance.

TABLE II
ABLATION STUDY OF THE EFFECTIVENESS OF BOTH SAR IMAGE AND TRIPLET LOSS (THE BEST RESULT IS SHOWN IN BOLD)

	MSE↓	SAM↓	PSNR↑	SSIM↑
w/o SAR nor triplet loss	0.000206	2.5116	43.1505	0.9532
w/o SAR	0.000213	2.3415	43.9486	0.9561
w/o triplet loss	0.000159	2.2671	44.0440	0.9617
TransGAN-CFR	**0.000149**	**2.2053**	**44.3781**	**0.9633**

2) Selection of FFN in the Encoder–Decoder Network: To verify our selection of FFN in the proposed framework, we tested three FFN variants with different modules. Concretely, *GD* represents the FFN architecture in a separate branch with a DepthConv block and GELU activation, *TG* represents the FFN architecture in two branches without DepthConv blocks, and *TD* represents the FFN architecture in two branches without GELU activation. The corresponding results are given in Table III. We can find that the dual-flow mechanism brings the most significant performance improvements among the three components since it enhances spatial information through calculations on double convolutional channels. DepthConv blocks and gating activation also contribute to performance improvements of the proposed method owing to their low-coupling operation mode and information filtering mechanism. Therefore, the design of FFN in this article is rational and effective.

3) Different Stage Numbers of the Encoder–Decoder Network: To explain the reason for the selection of the proposed encoder–decoder network structure, we tested five different generator structures with different hierarchical stage numbers. $K_S = 3$, $K_S = 4$, $K_S = 5$, and $K_S = 6$ stand for the generator structure with three, four, five, and six encoder/decoder stages, respectively. Moreover, the design of the last encoder stage was also tested. In the basic generator structure, a downsampling block is connected following every two encoding

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TABLE III
ABLATION STUDY OF THE GENERATOR (THE BEST RESULT IS SHOWN IN BOLD)

Network	Componet	MSE↓	SAM↓	PSNR↑	SSIM↑	FLOPS	Params
Feedforward Network	*GD*	0.000256	3.1713	41.2537	0.9370	49.41 GMac	58.05 M
	TG	0.000165	2.4054	43.6549	0.9594	19.04 GMac	74.43 M
	TD	0.000185	2.6015	43.1977	0.9536	20.99 GMac	74.43 M
Layer Number	$K_S = 3$	0.000289	3.2653	41.2299	0.9239	15.47 GMac	6.64 M
	$K_S = 4$	0.000174	2.3957	43.1971	0.9555	19.34 GMac	26.39 M
	$K_S = 5$	0.000155	2.2309	44.3552	0.9626	23.15 GMac	104.93 M
	$K_S = 6$	0.000196	2.5939	43.2163	0.9516	26.93 GMac	418.24 M
TransGAN-CFR		**0.000149**	**2.2053**	**44.3781**	**0.9633**	20.99 GMac	74.43 M

transformer block to double the feature dimension, and the parameters of upsampling blocks on the decoder side are determined according to the feature dimension from skip connections. In our TransGAN-CFR network, the downsampling blocks of the last encoder stage keep the feature dimension unchanged, while those of comparative structures still double the feature dimension. Table III shows the results of different generator structures.

As shown in Table III, the hierarchical structure of $K_S = 3$ is not sufficient to generate a complete reconstruction result with obvious content loss and color distortion in the generated images. With the increase in encoder–decoder stages, the extracted features are generally more abundant and diverse, so the reconstruction performance of the thick cloud areas becomes more reasonable and reliable. Note that in the structure of $K_S = 6$, the performance has decreased compared with the case $K_S = 5$, which reveals that the appropriate number of encoding and decoding stages is necessary. From the results of structure $K_S = 5$ and the proposed method, it can be disclosed that the strategy of maintaining the feature dimension at the last encoder stage avoids the generation of additional textures. According to the evaluation indicators displayed in Table III, the architecture of our method also shows the best performance.

V. CONCLUSION

In this work, we propose a transformer-based generative adversarial network through multisensor data fusion, called TransGAN-CFR, for cloud-free image reconstruction of satellite images. This method guides the generation of texture details of cloud-contaminated regions based on the complementary information embedding in SAR data. The generator framework is an elegant U-shaped encoder–decoder structure with transformer blocks to combine both the advantages of taking shallow–deep features and global–local context into account. Triplet loss is introduced to further ensure the quality of the generated image, which makes the generated image similar to the real cloud-free image but different from the real cloudy image. After the adversarial learning of the GAN, the best generator parameters are obtained. From the extensive experiments, our TransGAN-CFR model performs favorably and our reconstructed cloud-free images are visually better than the compared deep learning cloud removal methods. The effectiveness and reliability of our model are confirmed by ablation experiments.

In the future, there are still multitudinous research prospects and directions for cloud-free multispectral image reconstruction which can be pointed out as follows.

1) Hidden information in SAR data shall be dug and integrated to make the cloud removal network more interpretable.
2) Developing target models with stronger generalization capabilities is meaningful. Optical images often suffer from various degradations. Taking the physical mechanisms of various degradation problems into account may be beneficial for designing extensively used reconstruction models.
3) Given that the purpose of reconstructing cloud-free multispectral images is usually to achieve continuous observation for a certain region, research on cloud-free image reconstruction networks on time-series datasets is an effective means to achieve this goal. Moreover, training the network with time-series datasets may enable the reconstruction model to better deal with temporal changes caused by land cover, atmosphere, or lighting circumstances.
4) It is feasible to develop multitask models that include the cloud removal task as a front part.
5) Designing lightweight cloud removal networks with less computing cost is also a future research direction worthy of concentration.

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